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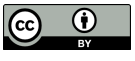
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Towards an Automatic Easy-to-Read Adaptation of Morphological Features in Spanish Texts

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Abstract. The Easy-to-Read (E2R) Methodology was created to improve the daily life of people with cognitive disabilities. This methodology aims to present clear and easily understood documents. The E2R Methodology includes, among others, a set of guidelines related to the writing of texts. Some of these guidelines focus on morphological features that may cause difficulties in reading comprehension. Examples of those guidelines are: (a) to avoid the use of adverbs ending in -mente (-ly in English), and (b) to avoid the use of superlative forms. Both linguistic structures are quite long, which is also related to another E2R guideline ("The use of long words should be avoided"). Currently, E2R guidelines are applied manually to create easy-to-read text materials. To help in such a manual process, our research line is focused on applying the E2R Methodology in Spanish texts in a (semi)-automatic fashion. Specifically, in this paper we present (a) the inclusive design approach for the development of E2R adaptation methods for avoiding adverbs ending in -mente and superlative forms, (b) the initial methods for adapting those morphological features to an E2R version, and (c) a preliminary user-based evaluation of the implementation of those methods.

Keywords: Easy-to-Read Methodology · Cognitive Accessibility · Artificial Intelligence

1 Introduction

People with cognitive disabilities present some difficulties related to reading comprehension processes. Hence, a methodology called Easy-to-Read (E2R) [1,13,17] was created with the goal of presenting clear and easily understood content. This methodology provides a collection of guidelines concerning both the content of texts and their design and layout, such as to use short and simple sentences, to avoid the use of long words, to divide ideas into paragraphs, or to use images that complement the content of the text.

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Currently, the E2R methodology is applied in a manual fashion. This adaptation process is iterative and involves three key activities: analysis, adaptation, and validation [1]. This manual process is labour-intensive and costly, and it would benefit from having a technological support. In this context, our research line is focused on applying different Artificial Intelligence (AI) methods and techniques[1] to (semi)-automatically perform the analysis and the adaptation of documents to obtain easy-to-read versions of original documents written in Spanish. Specifically, this paper is focused on two of the E2R guidelines that affect the writing of texts [1]: (a) to avoid the use of adverbs ending in -mente (-ly in English) and (b) to avoid the use of the superlative form of adjectives and adverbs. These two guidelines are considered useful for the daily work of E2R experts by one in every four experts as reported in [27].

In linguistic terms, these two structures are the result of the so-called process of word formation. Among other word classes, this process results on derived words, those formed from another word by the addition of morphological features called morphemes (e.g. *walk: walker*). Both the adverbs ending in -mente and the superlatives are examples of morphologically-derived words, as they are made up of a base attached to a morpheme (i.e. a suffix). With these notions in mind, it is important to note that morphological awareness, defined as the ability to recognise and manipulate morphemes [10], is related to the reading comprehension ability for different target groups including persons with cognitive impairments such as dyslexia (e.g. [25,26]). Research on this topic (see [26] for review), indeed, supports that poor comprehenders' difficulties have to do with derivational morphology, since the process of derivation can change the word class of the base (for instance, from an adjective to an adverb, e.g. *dulce* ('gentle'): *dulcemente* ('gently')). Therefore, these types of words are normally long and can clutter the text if used too often, since the more stimuli (in letter or syllable form) a text contains, the more cognitive processing the reader requires in the comprehension task [12]. Besides, one of the morphology-based linguistic accessibility criteria set by Jenge and colleagues [14] warns that morphologically complex words can impose a considerable burden to the human language processing system. Moreover, the reduction of morphological complexity of these features is related to another E2R guideline that recommends to avoid the use of long words, as they negatively affect language comprehension [4,6].

To improve the E2R adaptation process, technological aids could be developed. In this regard, to the best of our knowledge, research work on an automatic E2R analysis for Spanish is quite scarce. We can only mention an E2R conformance checker called Easy-to-Read Advisor [28]. Regarding E2R transformations for Spanish texts in an automatic way, it is worth mentioning Simplext [22], LexSIS [8], DysWebxia [20], and easier [16], which are based on simplification techniques. In such works, both lexical and syntactic simplification tasks are performed. However, none of the aforementioned works specifically addresses the E2R adaptation of either adverbs ending in -mente nor superlative forms.

To cover the identified gap, we pose the following research question: "Is it possible to develop an automatic method for adapting sentences written in

[1] We are investigating both the symbolic and subsymbolic approaches in AI.

Spanish that include both adverbs ending in -mente (-ly in English) and superlative forms into a simpler version, that is, an E2R version?" To answer our research question, we decided to follow an inclusive design approach [30], with the goal of involving a diverse set of stakeholders in our research. This approach implies that we conducted a user study to include people with cognitive disabilities in the activity of selecting the best proposal for adapting sentences with such morphological features (adverbs ending in -mente and superlatives). Based on these results, we have created declarative[2] methods for identifying and adapting these types of morphologically-derived words to the E2R Methodology and implement a pair of proofs of concept based on such methods. Since most of E2R experts prefer an application providing suggestions of E2R adaptations [27], when possible we create our methods based on different possible E2R adaptations. Finally, we performed a preliminary user-based evaluation of those proofs of concept.

The rest of the paper is organised as follows: Sect. 2 is devoted to the state of the art on (a) the most relevant linguistic features of adverbs in -mente and superlatives in Spanish, and (b) the automatic approaches for identifying and adapting these types of morphologically-derived words. In Sect. 3 we explain the user study we conducted with the aim of selecting the best proposal for adapting sentences with adverbs ending in -mente and/or superlatives. Section 4 presents our first attempts of methods for adapting both types of words to the E2R Methodology, the first versions of proofs of concept for those methods, and the summary of the preliminary user-based evaluation. In addition, the most essential drawbacks of the research presented in this paper are shown in Sect. 5. Finally, we present some conclusions and future work on this research.

2 State of the Art

As mentioned in Sect. 1, in this work we concentrate our efforts on developing initial methods to automatically detect derived words by morphological features such as adverbs ending in -mente and superlative forms in Spanish, and adapt them into easy-to-read versions, in the light of the guidelines provided by the E2R Methodology [1]. Thus, this section is devoted to (a) highlight some notes on both linguistic structures to better understand the problematic they raise (Sects. 2.1 and 2.3), and (b) summarise the automatic approaches for identifying and adapting such morphological features (Sects. 2.2 and 2.4).

2.1 Linguistic Features of Adverbs Ending in -mente

Adverbs ending in -mente belong to the so-called adverbs of manner. Formally, they are composed of feminine and singular adjectives (*lenta*: *lentamente*, 'slow: slowly'), or of single-ending adjectives (*feliz*: *felizmente*, 'happy: happily').

These adverbs conform a paradox, not always noticed in classical grammatical studies, but recognised directly or indirectly in modern ones [19]. Such a paradox

[2] Human knowledge is explicitly represented in a declarative form (e.g. facts and rules). This way to proceed is part of the so-called symbolic AI.

is a consequence of the fact that the noun *manera* ('manner') acquires a very abstract meaning in the paraphrase "de manera + adjective". Hence, when we say *Los votantes acudieron masivamente a las urnas*[3], we literally express a certain 'manera de acudir' ('manner of attendance'), but, at the same time, it is not evident that the property of 'ser masivo' ('to be massive') can express any manner itself [19].

In accordance with the Spanish Royal Academy [19], we distinguish among the following types of adverbs in -mente: (a) **Subject-oriented adverbs**, which refer to a certain property of an action, but also of the person or thing designated in the situation being described. Some examples of these adverbs are[4]: *deliberadamente, descuidadamente, or (in)conscientemente*. Moreover, (b) **Object-oriented adverbs** indicate the way in which the action affects the complement of some predicate. For instance, in *Cortó el pastel profundamente*[5], the adverb *profundamente* ('deeply') conveys indeed a way of slicing, but it mainly reports a certain change of state in the object which receives the cut, and not a situation of the subject who caused the deep cut. Following, (c) **Action-oriented adverbs** refer to certain obstacles that the action has to overcome, or to other circumstances that intrinsically characterise the action (e.g. *arduamente, dificultosamente, dolorosamente, or fácilmente*[6]). Finally, (d) **Adverbs of point of view or relational adverbs** are derived from relational adjectives. Since such adjectives establish a connection with a certain field or domain represented by the modified noun instead of denoting a property or quality, relational adverbs are hardly commutable by the paraphrase "de manera/forma + adjective".

2.2 Automatic Approaches for Identifying and Adapting Adverbs Ending in -mente

In Spanish the attempts on identifying and/or transforming this type of adverbs are way scarce. The automatic identification has been covered in some studies dealing with adverbs in general [21]; however, in the case of transformation, the way forward is still open. On the contrary, in other languages, such as French [29] or Portuguese [5], progress has been made along this way. On the one hand, the goal of Tolone and Voyatzi [29] is to extend the adverbial entries of LGLex, a Natural Language Processing (NLP) oriented syntactic resource for French. To do that, they first identify the adverbs ending in -mente, and then, for each type of adverb, they include the different paraphrasing alternatives in the lexical tables of LGLex. On the other hand, Baptista [5] aimed to provide a comprehensive set of paraphrasing strategies, which can be used in several natural language applications, such as text simplification or even machine translation. In this case, an annotated corpus was used to propose different paraphrases for each type of adverb, considering the lexical-semantic information. Then, they formalised such paraphrases through a finite-state automata to transform the equivalent adverb.

[3] Translation: *Voters went massively to the polls.*
[4] Translation: *deliberately, negligently, (un)consciously.*
[5] Translation: *He sliced the cake deeply.*
[6] Translation: *arduously, difficultly, painfully, easily.*

2.3 Linguistic Notions of Superlative Forms

Superlative is understood as the ponderation in a maximum or minimum degree of quantity or quality [18]. The superlative degree can be denoted by grammatical categories such as adjectives and adverbs. The expression of the superlative can be performed by means of numerous formulas [9]: (a) **Morphemic expression**, also called synthetic form, which is expressed through the process of affixation on adjective and adverbial bases, i.e. the use of prefixes such as "super- + base" (e.g. *supergrande* ('superlarge') and suffixes such as "base + -ísimo/a or -érrimo/a[7]" (e.g. *grandísimo* ('mighty') or *libérrimo* ('very free')). Another formula is the so-called (b) **Lexical expression**. In that case, the expression of the superlative occurs when a word carries itself the feature 'superlative', such as some quantifier adverbs (e.g. *demasiado* ('too much')), absolute adjectives (e.g. *excelente* ('excellent')), or adverbs ending in -mente, e.g. *completamente* ('completely')). For its part, the (c) **Syntagmatic expression**, also called analytic form, it is denoted by the adverb mark *muy* ('very') preceding an adjective or adverb (e.g. *muy triste* ('very sad')). Such a formula is the most common in Spanish [18]. Finally, superlative structures can be made up by some expressions as *un montón* or *miles de* (both mean 'a lot' and belong to a colloquial register of language), called (d) **Locutions or superlative expressions.**

Since the superlative structure affects the degree of the word it quantifies, it can only occur with words that are susceptible to gradation, which means that it must be possible to place them on a scale of comparison, at a position higher or lower than the one indicated by the adjective or adverb alone. In the case of adjectives, most of the so-called qualifying adjectives accept gradation, with some exceptions such as the adjectives of extreme degree (also called *elatives*) which correspond to the aforementioned lexical expression of superlative, since they express themselves the maximum degree of gradation, so they do not admit any type of affixation (e.g. *fabuloso: *fabulosísimo; enorme: *enormísimo*[8]). Due to their affective connotations, adjectives derived through this suffix (in this case, we refer only to the form -ísimo/a) are very rare in scientific and technical language, but very frequent in the colloquial language [19].

At last but not least, following Olmos [18] the morphemic expression using the suffixes -ísimo/a and -érrimo/a represent an exceptional paradigm in the word derivation process, due to the excessive length of the result (the suffixes

[7] Both suffixes express the same superlative feature. However, superlatives ending in -ísimo/a are widespread used in comparison to those in -érrimo/a. The suffixes -ísimo and -érrimo refer to the masculine gender, while -ísima and -érrima to the feminine gender. (Henceforth, we will use the slash symbol (/) to include both genders).

[8] In the colloquial register it is common to use the suffix -ísimo/a in elative adjectives to focus on the meaning of the adjective, so we can find this type of structures in spoken language, even though they are non-normative. (Henceforth, the asterisk symbol (*) will be used to indicate ungrammatical or non-normative structures).

add three syllables to the base) and the resulting proparoxytone[9] schema of the superlative, which can be a hard-to-pronounce word.

2.4 Automatic Approaches for Identifying and Adapting Superlatives

Although superlative forms have received considerable attention in formal linguistics [7], this interest is not mirrored in computational linguistics and NLP.

For such a reason, the study carried out in [7] is seen as the first automated approach to the interpretation of superlatives for open-domain texts in English. In such a research work, they present a corpus annotated for superlatives and propose an interpretation algorithm that uses a wide-coverage parser. The system they implemented is able to recognise a superlative expression and its comparison set. In addition, Jindal and Liu [15] studied the identification of superlatives in the framework of identifying comparative sentences in evaluative texts, and extracting comparative relations from them. To achieve such aims, they proposed two techniques to perform the tasks, based on class sequential rules and label sequential rules. Furthermore, it is worth mentioning the work led by Scheible [23], who proposed a computational treatment of superlatives aimed to automatically extract useful information from superlatives occurring in free text. Further on, the author extended this work in her doctoral thesis [24]. In the case of Spanish language, there are no automatic approaches for identifying superlatives and transforming them into simpler paraphrases.

3 E2R Adaptation Design via a User Study

As mentioned in Sect. 1, our intention is to involve a diverse set of stakeholders[10] in our research work. In this regard, we decided to apply an inclusive design approach [30] including people with cognitive disabilities in the team in charge of designing the most appropriate E2R adaptations for both adverbs ending in -mente and the so-called morphemic superlatives. Out of the scope of this paper, and thus one limitation of this research work, is the involvement of E2R experts in such an inclusive design approach.

To materialise the approach, we created an inclusive co-design process for selecting the most appropriate E2R adaptations. Thus, we developed a pair of questionnaires written in Spanish and implemented as a Google Form[11]. The goal of these questionnaires is to gather opinions of people with cognitive impairments on the use of adverbs in -mente and superlatives. Such questionnaires were

[9] Linguistic term for a word with stress on the antepenultimate (third last) syllable such as the words in English *cinema* and *operational*.

[10] The main actors who affect and/or are affected by our research line on cognitive accessibility are people with cognitive disabilities and E2R experts.

[11] Questionnaires are available at https://doi.org/10.5281/zenodo.8018593.

launched in February 2022 through mailing lists of autonomous federations and associations of people with cognitive disabilities in Spain[12].

3.1 User Study Design

Both questionnaires are divided into two main parts: (1) a section that includes single-answer multiple choice questions to capture data about which linguistic structures (adverb paraphrasing formulas in the case of adverbs ending in -mente and superlative forms in the other case) are easier or better understood; and (2) a part with questions on participants' demographics, knowledge, background, and experience. In the first part of the questionnaires, questions are of two different types. Such types of questions used in the questionnaire were validated by an E2R expert in a pilot survey before the user study began. On the one hand, the questionnaire poses questions where the participant has to choose an answer. In this case, participants are asked about their preferences or about the simplest answer. Possible answers to these questions consist of (a) sentences including original linguistic structures (i.e. adverbs in -mente or a superlative form that used a suffix (e.g. -ísimo/a)) and (b) one or more sentences that are the result of adapting the original linguistic structures with an E2R approach in mind, that is, trying to find a synonym formula which is easier to understand. On the other hand, there are questions where participants have to complete a sentence by selecting one of the possible answers. The set of answers includes the original linguistic structure and one or more synonym formulas. As an illustration, Table 1 shows two examples of these types of questions.

The collection of original sentences used in the questionnaire was built using two oral corpus: COSER[13] and C-Or-DiAL[14]. Synonym formulas were manually built by consulting several dictionaries such as WordReference[15] and Reverso[16]. A linguistic expert validated the collection of synonym formulas.

In the case of the questionnaire involving adverbs ending in -mente, the first part is composed of 16 questions of which four include sentences with subject-oriented adverbs, six include sentences with action-oriented adverbs, and six include sentences with adverbs of point of view or relational adverbs, based on the classification presented in Sect. 2.1.

While in the case of the questionnaire about superlatives, the first part is composed of 11 questions including morphemic superlative expressions (see Sect. 2.3). Out of these questions, nine contain sentences with superlative adjectives and two with superlative adverbs.

[12] Plena Inclusion España, an associative movement who fights in Spain for the rights of people with intellectual or development disabilities and their families, played the key role of (a) finding federations willing to participate in the user study and (b) distributing the information about this research work through those federations.

[13] https://hispanismo.cervantes.es/recursos/coser-corpus-oral-sonoro-del-espanol-rural.

[14] http://lablita.it/app/cordial/corpus.php.

[15] https://www.wordreference.com/.

[16] https://synonyms.reverso.net/sinonimo/.

Table 1. Sample questions from the questionnaires of both cases.

CASE 1: Adverbs ending in -mente
Question
Por favor, completa el principio de esta frase con una de las opciones: "..... Ana necesita más vocabulario técnico." (Translation (Tr.): *Please complete the beginning of this sentence with one of the options: "..... Ana needs more technical vocabulary".)*
Options
1. Definitivamente (Tr.: *Definitely*)
2. Sin lugar a dudas (Tr.: *Without doubt*)
3. Está claro que (Tr.: *It is clear that*)
4. En definitiva (Tr.: *In the end*)
5. No lo sé. (Tr.: *I do not know*)
6. Otro: (texto libre) (Tr.: *Other* (Participants can freely write their own proposals)).
CASE 2: Superlatives
Question
Por favor, lee las siguientes 2 frases. Qué frase te parece más sencilla? (Tr.: *Please read the next 2 sentences. Which sentence do you think is easier?*)
Options
1. David vive lejísimos de su prima Anabel. (Tr.: *David lives far away from his cousin Anabel*).
2. David vive muy lejos de su prima Anabel. (Tr.: *David lives very far away from his cousin Anabel*).
3. Niguna de las frases es sencilla. (Tr.: *None of the sentences is simple*).
4. Todas las frases son sencillas. (Tr.: *All sentences are simple*).
5. No lo sé. (Tr.: *I do not know*).
6. Otro: (texto libre) (Tr.: *Other* (Participants can freely write their own proposals)).

3.2 Participants in User Studies

On the one hand, for the survey on adverbs in -mente, 139 people responded the questionnaire (72 male, 60 female, and 7 participants who preferred not to provide gender information). Participants include representatives from five different autonomous communities in Spain[17]: Andalucía (73), Madrid (32), Comunidad Valenciana (31), one participant from Galicia and another one from Cataluña. Most of the participants (73.5%) answered the questionnaire alone, while the rest needed the support of another person. On the reading comprehension level[18], 51.1% of the participants had a medium level of reading comprehension, whereas 35.3% had a high level, 0.7% a medium-high level, and 5.8% a low level.[19]

Regarding the age range, half of the participants ranged from 31 to 45 years old, 25.9% from 18 to 30, 18.7% were from 46 to 60, two participants was over 60 years old, and 3.6% of the participants declined to provide their age. With

[17] Autonomic Federations from Andalucía, Comunidad Valenciana and Madrid played the crucial role of distributing the goal of this user study as well as the link to the questionnaire through different organisations of people with cognitive disabilities.

[18] The level of comprehension is based on a self-assessment question.

[19] 4.3% of the participants declined to provide their level, while 2.9% did not know it.

respect to their impairments, most of the participants (80.3%) had an intellectual disability, followed by those (8%) who had intellectual and physical disabilities. On the most frequent occupation of the participants, from the 139 participants, 68 were users of occupational centres, 10 were public examination candidates, 7 were unemployed, while 7 were E2R validators.

On the other hand, 121 participants (62 male, 54 female, and 5 participants who preferred not to provide gender information) took part in the survey on superlative forms. In this case, these participants came from seven different autonomous communities in Spain: Andalucía (68), Comunidad Valenciana (29), Madrid (20), one participant from Galicia, one from Cataluña, one from Castilla y León, and another one from Extremadura. Most of the participants (80.2%) answered the questionnaire alone, while the rest needed the support of another person. Regarding their occupation, 58.7% were occupational center users. With respect to participant's level of reading comprehension, 48.8% had a medium level of reading comprehension, while 0.8% had a very high level, 34.7% had a high level, 0.8% a medium-high level, and 9.9% a low level[20]. On the age range, around half of the participants (52.1%) ranged from 31 to 45 years old, 21.5% from 18 to 30, 20.7% were from 46 to 60, 2.5% were over 60 years old, 2.5% of the participants declined to provide their age and 0.8% provides a non-valid age. On their impairments, most of the participants (86%) had an intellectual disability, followed by those (6.6%) who had intellectual and physical disabilities.

3.3 User Study Outcomes: E2R Adaptation Proposals

Findings indicate that, overall, participants consider simpler those sentences that have been manually simplified by means of substituting the original linguistic structure (adverbs ending in -mente and superlatives) by other synonym formulas. In the questionnaire on adverbs ending in -mente, only in 3 out of 16 questions the preferred option was the one with the adverb ending in -mente. While in the case of questionnaire involving superlatives, in the 11 questions the preferred option was the one with a synonym paraphrasing of the original superlative form. The detail analysis of the data gathered is organised in Case 1 (Adverbs ending in -mente) and Case 2 (Superlatives).

Case 1: Adverbs ending in -mente. The situation in which the preferred option was the original sentence with an adverb ending in -mente includes, in particular, the following adverbs *directamente* ('directly'), *normalmente* ('normally') and *seriamente* ('seriously'). Our first hypothesis for explaining this outcome was that these three words would be very frequent in Spanish. In order to confirm such an hypothesis we analysed the frequencies of these three adverbs in the list of most frequent words in CORPES XXI (Corpus del Español del Siglo XXI)[21]. However, these adverbs were not in the top of frequencies; indeed, other

[20] 1.7% of the participants declined to provide their level, while 0.8% did not know about their level and 2.5% did not know how to read.

[21] https://www.rae.es/banco-de-datos/corpes-xxi.

adverbs in our questionnaire have highest frequencies (e.g. *actualmente* ('currently') and *solamente* ('only')). Nevertheless, in very broad terms, we could interpret this situation from a morphological point of view, since these three preferred adverbs (*directamente* ('directly'), *normalmente* ('normally') and *seriamente* ('seriously')) belong to the group of relational adverbs, and thus they are made up by relational adjectives, as we mentioned in Sect. 2.1. This type of adjectives do not express a single property or quality of the noun they accompany, but denote a set of properties and link them to those of the modified noun, thus they establish different types of more complex semantic relations [11]. In addition, the constituent -mente neither alters the semantics of the adjective nor changes its category, hence the adjective still exhibits all the formal properties that it exhibited as an independent adjective [2]. This means that, for example, in the paraphrase "de manera + adjective" the adjectives *seria*, *normal* and *directa* do not qualify the property of the noun *manera*, but express a set of properties that affect the whole sentence. For such a reason, the replacement of those three adverbs (*directamente, normalmente* and *seriamente*) by the paraphrase "de manera + relational adjective" may seem unusual for persons with reading comprehension difficulties.

A more in depth analysis[22] of the survey data reveals that participants' preferred selections can be classified into the following scenarios:

- **Scenario A.** Sentences that are adapted by using the following pattern "de forma + adjective", e.g. original sentence: *Antonio escribe correctamente, sin faltas de ortografía*[23]; adapted sentence: *Antonio escribe de forma correcta, sin faltas de ortografía*[24]. Two survey responses fall into this category.
- **Scenario B.** Sentences that are adapted by replacing the adverb ending in -mente by synonym paraphrasing structure, e.g. original sentence: *Probablemente los pintores necesiten más botes de pintura*[25]; adapted sentence: *Puede que los pintores necesiten más botes de pintura*[26]. Six responses fall into this grouping.
- **Scenario C.** Sentences that are adapted by eliminating the adverb ending in -mente, e.g. original sentence: *María asiste a clase de español solamente un día por semana*[27]; adapted sentence: *María asiste a clase de español Ø un día por semana*[28]. Two survey responses fall into this category.
- **Scenario D.** Sentences that are adapted by replacing the adverb ending in -mente by a synonym word, e.g. original sentence: *Antiguamente, los niños entraban en la escuela con 6 años*[29]; adapted sentence: *Antes, los niños entraban en*

[22] The analysis and aggregation of ratings was performed manually grouping the different percentages of response types (no statistical analysis tool was used). Data gathered from the questionnaires are available at https://doi.org/10.5281/zenodo.8018593.

[23] Translation: *Antonio writes correctly, without spelling mistakes.*

[24] Translation: *Antonio writes in a correct way, without spelling mistakes.*

[25] Translation: *The painters will probably need more paint cans.*

[26] Translation: *Painters may need more cans of paint.*

[27] Translation: *María attends Spanish classes only one day a week.*

[28] Translation: *María attends Spanish classes one day a week.*

[29] Translation: *Formerly, children entered school at the age of 6.*

la escuela con 6 años[30]. Three survey responses fall into this group. In this case, it is worth mentioning that the sentence with the adverb ending in -mente was the third preferred option in all the questions.

In addition, we analysed the data gathered for each scenario with respect to the following variables: age, level of comprehension and origin. Table 2 shows the corresponding percentages[31].

Table 2. Summary of percentages for each scenario in Case 1.

Case 1: Adverbs ending in -mente					
Variable		Scenario A	Scenario B	Scenario C	Scenario D
Age	**18–30**	9%	47%	4%	12%
	31–45	12%	45%	6%	12%
	45–60	9%	49%	10%	18%
Reading Comprehension	**High**	22%	44%	3%	16%
	Medium	18%	49%	11%	13%
	Low	9%	42%	10%	13%
Geographical Origin	**Madrid**	11%	39%	6%	14%
	Andalucía	12%	41%	7%	10%
	Comunidad Valenciana	9%	45%	8%	12%

Figure 1 shows which percentage of participants selected the most preferred option and what percentage the sentence with adverb ending in -mente in the four identified scenarios. For each scenario, it is shown the percentage of answers of the participants who have chosen the sentence with the adverb in -mente, or the sentence with the adapted paraphrase. As we can see in the figure, in all the scenarios there is a clear difference between the percentage of participants who selected the most preferred option and those who selected the sentence with the adverb in -mente. In fact, the percentages for the most preferred options are in almost all cases more than double the percentages for the sentences with adverbs in -mente. As a summary, the situation represented by Scenario B has been the one that has appeared the most, followed by the situation described in Scenario D. Finally, the situations represented by Scenarios A and C have been the ones that have occurred the least.

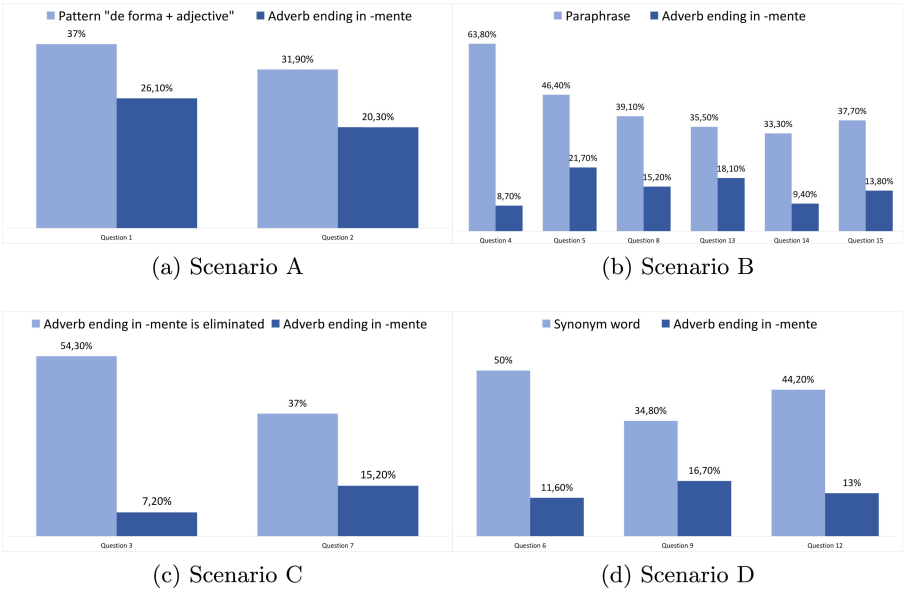
Case 2: Superlatives. A deeper analysis of the survey data reveals that participants' preferred selections (that is, those options with the highest percentage) can be classified into the following scenarios:

- **Scenario A.** Sentences that are adapted by directly using the original adjective or adverb, without including any mark for its gradation, e.g. original sentence: *El suegro de Alba está contentísimo*[32] and the preferred adapted sentence: *El suegro*

[30] Translation: *Earlier, children entered school at the age of 6.*

[31] The regions of Galicia and Cataluña have not been considered as there is only one participant for each.

[32] Translation: *Alba's father-in-law is very happy.*



(a) Scenario A (b) Scenario B

(c) Scenario C (d) Scenario D

Fig. 1. Percentage of selections for the most preferred option from the adapted formulas and for the sentence with adverb in -mente in the four scenarios.

de Alba está contento[33]. Five survey responses fall into this category. In this scenario, it is worth noting that we have identified two sub-scenarios regarding the participants' second preferred option: (1) **Sub-scenario A.1.** Synonym of the original adjective or adverb is used as an E2R paraphrasing. For instance, *feliz* ('happy') is a common synonym for *contento* ('pleased'). Three survey responses fit this category; and (2) **Sub-scenario A.2.** The original adjective or adverb including the gradation mark *muy* ('very') is used as E2R paraphrasing. For example, *rarísimo* ('strange') is substituted by *muy raro* ('very strange'). Two survey responses fall into this category. Given that, this Scenario A indicates the participants' preference in selecting the lexical expression of superlative when (a) an extreme-degree adjective is proposed as synonym of the morphemic expression ending in -ísimo/a (e.g. most participants prefer the extreme-degree synonym adjective *feliz* ('happy') instead of the morphemic form *contentísimo* ('very pleased')); and (b) the morphemic expression with the suffix -ísimo/a is added to a extreme-degree adjective or adverbial base (e.g. most participants prefer the original adjective *enojada* ('angry') instead of its morphemic version of superlative *enojadísima* ('very angry'), since *enojada* carries itself the maximum degree of superlative). Hence, we can find that participants consider easier or better understood the lexical expression of the superlative rather than its morphemic version with the suffix -ísimo/a.

[33] Translation: *Alba's father-in-law is happy.*

- **Scenario B.** Sentences that are adapted by replacing the original superlative by the original adjective or adverb, including the gradation mark *muy* ('very'), e.g. the original sentence: *Ana es guapísima*[34]; and the preferred adapted sentence: *Ana es muy guapa*[35]. Six questions fall into this grouping. In such a scenario, we observed that those six cases correspond to the so-called restrictive qualifying adjectives, which point out characteristics that distinguish nouns among their peers, i.e. for instance, in the previous example *Ana es muy guapa*, the adjective *guapa* is "differentiating" Ana from the rest.

Although both the morphemic expression (suffix -ísimo/a and -érrimo/a) and the syntagmatic expression (formula "*muy* + adjective or adverb") are considered synonymous structures for expressing superlative, there are semantic differences in their use. According to [18], the synthetic form with -ísimo/a expresses a higher gradation than the analytic formula "*muy* + adjective or adverbial base". However, thanks to the findings extracted in this first approach, people with reading comprehension difficulties lean towards the latter form using *muy*, either due to the phonetic and morphological difficulties posed by the suffix -ísimo mentioned in Sect. 2.3, or due to the common use of this formula in Spanish.

In addition, we analysed the data gathered for each scenario with respect to the following variables: age, level of comprehension and origin. Table 3 shows the corresponding percentages[36].

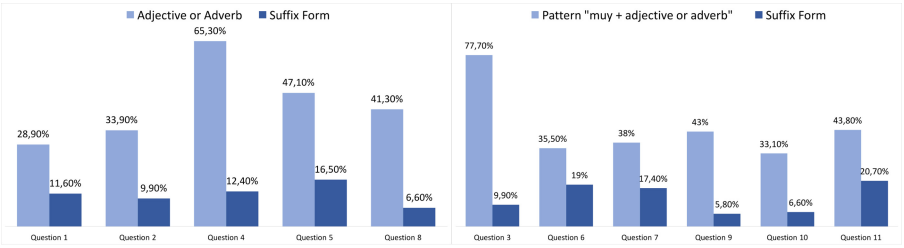
Table 3. Summary of percentages for each scenario in Case 2.

Case 2: Superlatives			
Variable		Scenario A	Scenario B
Age	**18–30**	31%	49%
	31–45	29%	51%
	46–60	35%	47%
	Over 60	30%	50%
Reading Comprehension	**High**	30%	48%
	Medium	35%	49%
	Low	31%	48%
Geographical Origin	**Madrid**	35%	50%
	Andalucía	30%	48%
	Comunidad Valenciana	32%	50%

Figure 2 presents which percentage of participants selected the most preferred option for those responses that fall into Scenario A (five responses) and Scenario

[34] Translation: *Ana is gorgeous.*
[35] Translation: *Ana is very pretty.*
[36] The regions of Galicia, Cataluña, Castilla y León, and Extremadura have not been considered as there is only one participant for each.

B (six responses), respectively. For each scenario, it is shown the percentage of answers of the participants who have chosen the sentence with the suffix form in -ísimo/a, or the sentence with the adapted paraphrase. As can be seen in the figure, there is a clear difference between the percentage of participants who selected as the most preferred option the superlative paraphrasing (original adjective or adverb, with or without the gradation mark *muy* ('very')) and those who selected the sentence with the superlative form using a suffix. In fact, the percentages for the most preferred options are in almost all the cases more than double the percentages for the sentences with superlative features.



(a) Scenario A (b) Scenario B

Fig. 2. Percentage of selections for the most preferred adapted paraphrase and for the sentence with the superlative (using the suffix form) in the two scenarios.

Finally, it is worth mentioning that considering the data shown in Tables 2 and 3, there is no evidence that participants' responses are influenced (or mediated) by any of the categorical variables used to describe the sample (age, level of reading comprehension and region of origin), as the percentages are similar for each scenario in the three variables.

4 Initial Methods for an E2R Adaptation of Morphological Features

The general aims of the proposed methods are (a) to detect a couple of derived words by morphological features in texts written in Spanish, and (b) to provide the most appropriate E2R paraphrasing formulas for such linguistic structures. The selected paraphrasing formulas are based on the data gathered from people with cognitive disabilities during the inclusive design action explained in Sect. 3. Our initial methods are composed of the following activities: (1) NLP, which includes a cleanup of the text using regular expressions and a tokenization step, (2) Morphological Feature Identification, and (3) Morphological Feature Adaptation. The following sections explain the E2R adaptation method for adverbs ending in -mente (Sect. 4.1) and the method for adapting morphemic superlatives

ending in -ísimo/a or -érrimo/a (Sect. 4.2) as well as the services[37] that implement such methods. Finally, Sect. 4.3 shows our initial efforts for evaluating the implemented methods.

4.1 Method for Adapting Adverbs Ending in -mente

Adverbs identification activity relies first on a pre-filter of the words in the text in order to obtain only those words ending in -mente and second on a selection of the words that are adverbs from this pre-filtered set. The activity of adapting adverbs was conceived as an activity to provide first the most appropriate substitution and, second, an ordered list with other possible substitutions. This design decision was based on the E2R experts preferences about having adaptation suggestions in a support application [27]. All possible adaptations are based on the data gathered from people with cognitive disabilities described in Sect. 3.3. As outcomes of those data we obtained the following ranking of preferences for adapting adverbs in -mente: (1) using a synonym paraphrasing structure, (2) using a synonym word, (3) using the pattern "de forma + adjective", and (4) eliminating the adverb in -mente. Based on these outcomes, our initial method proposes as first adaptation the option described in Scenario B, i.e. a synonym paraphrasing structure. This was the first option because it is the most recurrent situation as shown in Sect. 3. The ordered list of possible substitutions is created with (1) the option presented in Scenario D (synonym word), (2) the option described in Scenario A (pattern "de forma + adjective", and (3) the option in Scenario C (deleting the adverb). The first two adaptation formulas imply the automatic identification of synonyms (a non trivial task).

We have developed a proof of concept, as a RESTful web service, to detect adverbs ending in -mente in Spanish texts and providing the most appropriate E2R adaptation formulas, based on the aforementioned method. Such a service requires as input a text written in Spanish and provides as output a set of possible E2R adaptations for adverbs ending in -mente in the input text.

The developed service has been implemented in Python 3.9, using the development framework Flask. This services uses LibrAIry[38] [3] for detecting adverbs ending in -mente. In particular, our service uses the following LibrAIry functionalities: Part-of-Speech (PoS) tagging and stemming. As a first attempt the method was implemented using a declarative mapping catalogue with adverbs in -mente and possible synonym paraphrasing structures or synonym words. This catalogue was used to obtain the first two adaptation formulas described in our method. We realised that this solution is not scalable, thus we decided to plan as a future activity the research of a more flexible solution. For this reason, currently, the service implements as possible E2R adaptation formulas those represented by Scenarios A (pattern "de forma + adjective", and C (deletion of the adverb). In addition, after discussing with a linguistic expert, we decided to include an additional pattern for Scenario A, that is, "de manera + adjective".

[37] Services are not currently available due to privacy constrains in the context of the project in which they has been developed.

[38] http://librairy.linkeddata.es/nlp/api.html.

We tested the identification functionality of our service with a collection of 2502 texts written in Spanish extracting from minutes of sessions held in municipalities[39]. We manually created a set of 2502 unit tests for testing the identification of adverbs ending in -mente. Each test is composed of the text written in Spanish and the annotation. This annotation contains the adverbs ending in -mente that appears in the text and was created in a manual fashion. Then, we developed an ad hoc testing code, written in Python, for (a) using our service for identifying adverbs ending in -mente in the text that is included in each unit test and (b) comparing the output of the service with the annotation included in the unit tests. After the testing activity, our service passed 2433 unit tests. Based on this, we can mention that our service identify adverbs ending in -mente with a 97% of success. An ongoing work is the design of evaluating activity for the adaptation functionality of our service.

4.2 Method for Adapting Superlatives

As in the case of adverbs ending in -mente, the activity of adapting superlatives was conceived as an activity for providing first the most appropriate substitution and, second, an ordered list with other possible substitutions. The idea is to have adaptation suggestions as preferred by E2R experts [27]. All possible substitutions are based on the data gathered from people with cognitive disabilities described in Sect. 3.3. As outcomes of those data we obtained the following ranking of preferences for adapting superlatives: (1) using the pattern "muy + adjective/adverb" and (2) using the adjective or adverb without gradation mark. Based on these outcomes, our initial method proposes as main adaptation the most recurrent situation shown in Sect. 3.3; that is, the option described in Scenario B (to use "muy + adjective/adverb" as a synonym paraphrasing structure). The ordered list of possible substitutions is created with (1) the option presented in Scenario A, that is, using the original adjective or adverb, without including any mark for its gradation; and (2) a synonym of the original adjective or adverb (Sub-scenario A.1). We considered interesting and useful to include an additional adaptation to have a more rich list of suggested adaptations. After selecting the most appropriate substitution, a post-processing should be performed in order to maintain the syntactic coherence in the sentence.

We have developed a proof of concept, as a RESTful web service, based on the aforementioned method. Such a service requires as input a sentence written in Spanish and provides as output a set of possible E2R adaptations for superlatives appearing in the input text.

The developed service has been implemented in Python 3.9, using the development framework Flask. This service uses spaCy[40] for sentence tokenization and TextServer[41] for tagging the obtained tokens with EAGLE PoS tags[42] The

[39] https://ayuntamientoboadilladelmonte.org/tu-ayuntamiento/gobierno-municipal/los-plenos/actas-municipales.
[40] https://spacy.io/.
[41] http://textserver.cs.upc.edu/textserver/login?hl=es.
[42] https://www.cs.upc.edu/~nlp/tools/parole-sp.html.

superlatives detection is based on a filtered process using the following PoS tags: category A (adjective) and degree S (superlative). When the superlatives have been identified, TextServer is used again in order to get the gender and number of such superlatives. This information is needed to maintain the syntactic coherence in the adapted sentence. Regarding the adaptation formulas, the service only implements as possible E2R adaptation formulas the ones represented by Scenarios B (pattern "muy + adjective/adverb") and A (adjective or adverb without gradation mark). Since the situation represented by sub-scenario A.1 implies the automatic identification of synonyms, which is not a trivial task, we decided not to implement in our service the option of adapting the superlatives by means of using a synonym of the original adjective or adverb.

We tested the identification functionality of our service with a collection of 566 sentences written in Spanish and extracted randomly from CREA Corpus[43]. We manually created a set of 566 unit tests for testing the identification of superlatives. Each test is composed of the text written in Spanish and the testing annotation. This annotation contains the superlatives that appears in the text and was created in a manual fashion. For performing the testing activity, we developed an ad hoc testing code, written in Python, for (a) using our service for identifying superlatives in the text that is included in each unit test and (b) comparing the output of the service with the testing annotation included in the unit tests. After the testing activity, our service passed 540 unit tests. Based on this, we can mention that our service identifies superlatives with a 95% of success. At present, we are designing the approach for evaluating the adaptation functionality of our service.

4.3 Preliminary Evaluation of E2R Adaptation Services

To evaluate the E2R adaptation services described in Sects. 4.1 and 4.2, we decided to perform an initial user-based validation via online questionnaires. Two validation settings have been considered: one devoted to the case of adverbs ending in -mente (Case 1) and the other to the use of superlatives (Case 2). For each setting, the questionnaire[44] was divided into two main parts: (1) one part with questions related to participants' demographics, knowledge, background and experience; and (2) the other part that includes 10 single-answer multiple choice questions to capture participants' opinions about how easy is the adaptation provided by our service. The format of these 10 questions is always the same. Each question has 2 sentences (one is an original sentence and the other one is the sentence adapted by our service). After reading each question calmly, participants should choose one option among the following ones: (a) Sentence 1 is the one you understand best; (b) Sentence 2 is the one you understand best; (c) I understand both sentences well; and (d) I do not understand either of the two sentences. The two questionnaires were validated by an E2R expert before

[43] https://www.rae.es/banco-de-datos/crea/crea-version-anotada.

[44] Questionnaires are written in Spanish, implemented as a Google Form, and available at https://doi.org/10.5281/zenodo.8018593.

starting the evaluation activity. 17 participants responded each questionnaire, 7 males and 10 females, all from Madrid. 13 of them had a medium level of reading comprehension and 4 a low level. Their distribution by age is 5, 3 and 9 participants in the ranges 18–30, 31–45 and 46–60 respectively. Regarding the impairments, 14 had an intellectual disability, one person had intellectual plus physical disability, one other had intellectual plus mental disability and the last one declared to have neurological issues. As for their occupation, 10 were users of occupational centres, 2 were E2R validators and the remaining were a cognitive accessibility professional, an unemployed person, a retired person, a receptionist, and a kit man.

Table 4. Summary of percentages in the user-based evaluation.

		Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
Original	Case 1	6	6	12	12	6	18	6	0	6	35
	Case 2	12	18	6	6	6	24	6	12	6	6
Adapted	Case 1	76	70	70	64	82	64	82	94	88	41
	Case 2	59	59	88	76	82	76	76	53	59	65
Both	Case 1	18	18	12	12	12	12	12	6	6	0
	Case 2	29	23	6	12	12	0	6	35	35	29
None	Case 1	0	6	6	12	0	6	0	0	0	24
	Case 2	0	0	0	6	0	0	12	0	0	0

Table 4 shows the percentages of participants who choose each of the available options for both settings (Case 1 and Case 2), i.e. choosing whether their preferences in terms of understanding were the original sentence (Original), the E2R adaptation (Adapted), both sentences (Both) or none of them (None). Analysing the first setting (Case 1), percentages clearly show their preference for the adapted text; they manifest it is the easiest to understand in almost the whole set of questions. Sentences in questions 1, 5, 7, 8 and 9 were understood by every participant. The percentage of them who prefer the adapted version is over 75%. Specifically, for question 8, all participants chose the adapted form except one who chose both forms. Question number 10 is the one furthest from this pattern, as it seems the hardest to understand. Despite this fact, the E2R adaptation is still the preferred one. When analysing the second setting (Case 2), again, the E2R adaptation is the preferred one. In this case, the comprehension of questions 1, 8, 9 and 10 is less impacted by the adaptation as the percentages of people who chose they understand both sentences are the highest. Yet, these percentages are always lower than the ones for the E2R option. Compared to Case 1, it seems the impact of the adapted forms in helping to understand superlatives is higher than the impact in adverbs in -mente. This is supported by the fact that 8 out of 10 questions in Case 2 have no observations in category

None, meaning everyone ended up understanding the sentence and preferring the E2R form.

Overall, these figures in Table 4[45] show the usefulness in terms of understanding of the E2R adaptation carried out by the developed E2R adaptation services.

5 Discussion

Regarding the user study described in Sect. 3, what is currently missing is to include people without cognitive disabilities, in particular, E2R experts in the design of the most appropriate E2R adaptations for adverbs ending in -mente and for superlatives. The opinion of E2R experts would be a very useful complement to the results obtained from people with cognitive disabilities. In this sense, the E2R adaptation design would include the visions of the two main actors who are interested in our research work.

Concerning the proposed methods for adapting adverbs in -mente and superlatives, we considered as E2R adaptation results the ranking of preferences obtained in the user study described in Sect. 3. These results were based on two surveys with 139 participants for the case of adverbs in -mente and 121 participants for the case of superlatives. However, those methods have been implemented as proofs of concept with slight adjustments in order to have more scalable services. Thus, this difference between the methods and the services is the main limitation of this part of our contribution. More intensive research is needed to find the best solution to have methods and services describing the same situations and providing the same results. In addition, we need to evaluate in a more exhaustive way both the identification and adaptation in our services.

Finally, with respect to the evaluation of the automatic methods described in Sect. 4, we performed an initial user-based validation of the services for adapting to an easy-to-read version adverbs ending in -mente and superlatives. The validation involved 17 participants. Results of this validation provides us with brief insights about the benefits of having adaptation services for these linguistic structures. The data analysis of this validation activity suggests that the service outcomes, in general, are easy to understand by people with cognitive disabilities, thus, such outcomes improve the understanding of people with cognitive disabilities. However, we should confirm these conclusions by means of performing a larger user-based validation that involves a higher quantity of people with cognitive disabilities.

6 Conclusions and Future Work

With the general aim of improving the cognitive accessibility of texts written in Spanish, in this paper, we present a pair of straightforward declarative methods

[45] Data gathered from the questionnaires as well as reports generated by Google Forms are available at https://doi.org/10.5281/zenodo.8018593.

for adapting sentences written in Spanish which include morphological features of derived words, such as adverbs ending in -mente and superlative forms, following an E2R approach. Although the initial methods we propose for the E2R identification and adaptation of adverbs ending in -mente and superlative forms are simple and direct, we consider them as a useful contribution in the context of cognitive accessibility and assistive technologies. Based on our service development, we can respond in a positive manner to the research question presented in Sect. 1 ("Is it possible to develop automatic methods for adapting sentences written in Spanish that include both adverbs ending in -mente (-ly in English) and superlative forms into a simpler version, that is, an E2R version?"). A crucial task in the development of those methods has been the selection of the most appropriate E2R paraphrasing formulas for such morphologically-derived words. To perform this task we applied an inclusive design approach involving people with cognitive disabilities. Their involvement was materialize by participating in two surveys. In those surveys participants were asked about their preferences with respect to the simplicity of a collection of short sentences written in Spanish. Thanks to the data collected in such questionnaires, we discovered the patterns to be used in the E2R adaptation of adverbs ending in -mente and superlatives in Spanish. After analysing those patterns, we realised that involving E2R experts in this inclusive design approach would have provided us with a more exhaustive vision of the E2R adaptation options.

The proposed methods have been implemented, with slight changes, as proof of concepts in the form of services. Such services has been initially evaluated by means of involving people of cognitive disabilities. This user-based evaluation has been performed by means of two questionnaires. This is a first approximation to a deeper evaluation of the services since the sample size was not large, but prospects are optimistic. Thus, currently, we could say that adaptations obtained from our services seem to be easier than original texts.

As further research, we have planned three specific directions: one pragmatical, one more analytical and another one more technical. First of all, we are going to organise a working session with E2R experts with the goal of complementing the data gathered from people with cognitive disabilities about the most convenient ways to adapt adverbs ending in -mente and superlatives to an easy-to-read form. Second, we are going to analyse in more depth the data gathered in our inclusive co-design process for selecting the most appropriate E2R adaptations for adverbs ending in -mente and superlatives. We would like to find linguistic patterns to explain the selection made by participants with respect to the types of such morphological features and the identified scenarios. And third, several technical actions are planned: (a) we are going to design the evaluation activity for the adaptation functionality of our services; (b) we are going to investigate different ways to obtain, in an automatic fashion, the equivalences between original morphological features and possible synonym paraphrasing structures; and (c) we have plan to integrate our services in a web user interface. After these technical improvements, we plan to perform a set of user-based studies to eval-

uate both the web user interface and the services. The plan is to involve both E2R experts and people with cognitive disabilities in these studies.

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